Wazuh Configuration, Attack Simulation & Log Monitoring

Submitted by: Athira Krishnan

Course: Certified IT Infrastructure & Cyber SOC Analyst (CICSA)

Acknowledgement

I would like to express my sincere gratitude to RedTeam Academy for giving me the opportunity to complete this project as part of the CICSA certification program. Working on this project helped me gain hands-on experience and a clearer understanding of how a Security Operations Center (SOC) functions in a real-world environment.

I would also like to my gratitude to the instructors and mentors for their consistent support, practical guidance, and constructive feedback throughout the course. Their direction played an important role in helping me complete this project successfully.

In addition, I am thankful for the resources and learning environment provided, which allowed me to strengthen my knowledge in cybersecurity monitoring, threat detection, and incident response.

Finally, I extend my gratitude to each individual who supported me, directly or indirectly, in achieving the successful completion of this project.

Athira Krishnan

Abstract

This project explains how to install and configure a basic Security Operations Center (SOC) lab using Wazuh on a Linux system. The objective was to build a centralized log monitoring and threat detection setup, perform controlled attack simulations, and observe how security events are detected and displayed through a real-time dashboard.

The lab was set up in a virtual environment. I used Ubuntu to configure the Wazuh Manager and Agent, and Kali Linux as the attacker machine. After installing and configuring the required Wazuh components, I enabled log collection and rule-based detection. The dashboard interface allowed me to monitor events and alerts as they happened.

I carried out different attack simulations, including SSH brute-force attempts, reconnaissance, privilege escalation activities and file modification attacks. These actions produced real security logs, which were collected and analyzed by the Wazuh Manager. Based on predefined rules, alerts were triggered and categorized by severity levels, helping us clearly understand how suspicious activities are identified and prioritized.

Through this project, I gained practical exposure to SIEM deployment, event correlation, and continuous security monitoring. It helped me understand how centralized monitoring plays an important role in detecting unauthorized access, tracking authentication failures, and responding to potential cybersecurity threats in real-world environments.

Contents

1. Introduction

A growing number of cybersecurity threats, including brute force attacks, malware infections, ransomware attacks, phishing and social engineering attacks, Distributed Denial-of-Service (DDoS) attacks, data breaches, zero-day exploits, web application attacks, and advanced persistent threats (APTs), are encountered by organizations and individuals in today's digital environment. Organizations rely on Security Information and Event Management (SIEM) systems and organized Security Operations Center (SOC) environments to efficiently monitor, identify, and address such threats. In order to reduce possible risks before they become serious security incidents, a SOC is essential for continuously evaluating system activity, connecting security events, and producing alerts.

The goal of this project is to leverage Wazuh, an open-source SIEM and host-based intrusion detection technology, to construct a simple SOC environment. The project's main goals are to set up Wazuh on a Linux-based system, simulate actual cyberattacks in a lab setting, gather and examine security logs, generate alerts based on detection rules, and visualize security events via a centralized dashboard.

A virtualized system including an Ubuntu server set up as the Wazuh Manager and another ubuntu as the Wazuh Agent and a Kali Linux computer used to conduct controlled attack simulations was used for the practical implementation. Through this project, key cybersecurity concepts such as log analysis, threat detection, rule-based alerting, and event correlation were explored.

This hands-on approach provides foundational knowledge of SOC operations and demonstrates the importance of continuous monitoring in maintaining system security and integrity.

2. Overview of Wazuh

Wazuh is a free and open-source security platform that unifies XDR and SIEM capabilities. It protects workloads across on-premises, virtualized, containerized, and cloud-based environments. Wazuh helps organizations and individuals to protect their data assets against security threats. It is widely used by thousands of organizations worldwide, from small businesses to large enterprises.

The Wazuh architecture components:

- Wazuh Manager (Server):
  The central component responsible for receiving logs from agents, analyzing events using predefined rules, correlating security data, and generating alerts.
- Wazuh Agent:
  Installed on monitored endpoints, the agent collects system logs, authentication logs, and security-related events, then securely forwards them to the Wazuh Manager for analysis.

- Wazuh Indexer:
  The indexer stores and indexes security event data, enabling efficient search, filtering, and retrieval of logs.
- Wazuh Dashboard:
  A web-based interface that allows users to visualize alerts, monitor security events, analyze trends, and manage agents in real time.

Together, these components create a centralized monitoring system that enables effective detection and analysis of cybersecurity threats within an organization**.**

3. Objectives

This project's main goal is to use Wazuh to create and implement a simple Security Operations Center (SOC) system in a controlled laboratory setting. The project's goal is to give participants hands-on experience with security monitoring, log management, and threat detection techniques by analyzing system activity in real time.

The specific objectives of this project are:

1. To install and configure Wazuh Manager and Agent on Linux-based systems.
2. To establish a centralized log collection and monitoring mechanism.
3. To simulate controlled cyberattacks.
4. To analyze security logs generated during attack simulations.
5. To generate alerts based on predefined detection rules.
6. To visualize security events and alert trends using the Wazuh Dashboard.
7. To understand the practical workflow of a Security Operations Center, including event correlation and incident monitoring.

4. Prerequisites

3.1 Hardware Requirements

- Minimum 8 GB RAM & 6 processors (Wazuh Manager)
- At least 50 - 70GB free disk space
- 2 - 4 GB RAM & 2 processors (Wazuh Agent and attacker Machine)

3.2 Software Requirements

- Ubuntu 24.04 LTS – Installed as Wazuh Manager
- Ubuntu 22.04 LTS – Installed as Wazuh Agent
- Kali Linux – Used as attacker machine
- VirtualBox – Virtualization platform
- Wazuh 4.14
- Hydra – For SSH brute force simulation

3.3 Network and Configuration Requirements

- NAT (Internet access for updates and installations)
- Host-Only (Internal secure communication between lab machines)

5. Procedures

This section describes the technical steps followed to deploy and evaluate a Security Operations Center (SOC) environment using Wazuh. The implementation was carried out in a virtual lab setup, including installation of the Wazuh Manager and Agent, network configuration, attack simulation, log collection, alert analysis and dashboard creation. The structured approach ensured proper system integration and effective monitoring of security events.

5.1 Environment Preparation

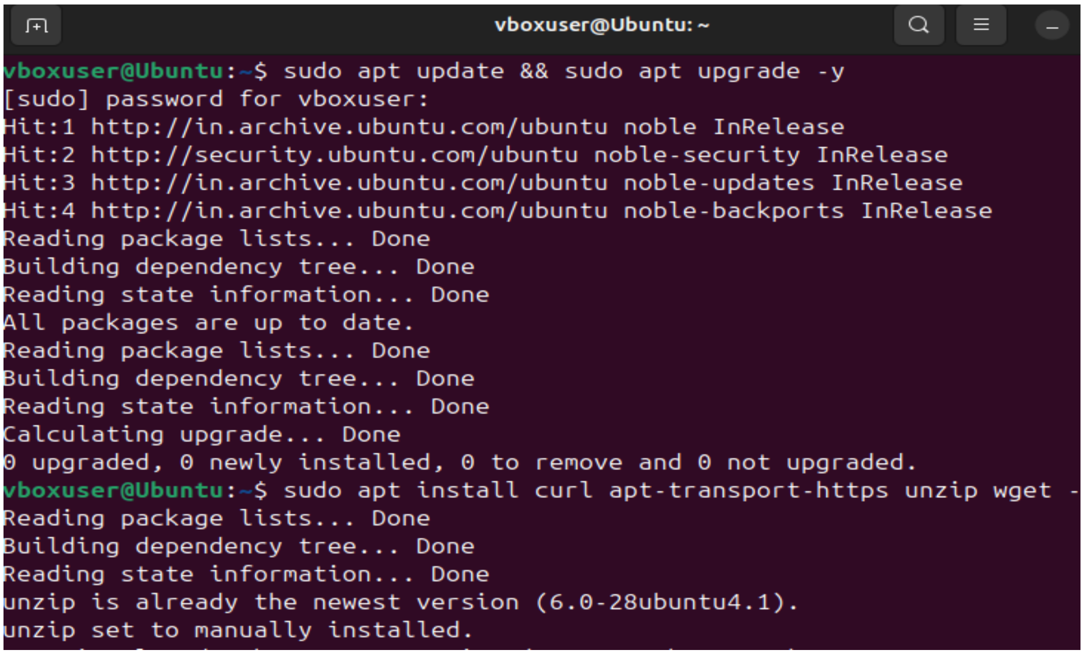
After installing VirtualBox as the virtualization platform, three virtual machines were created: Kali Linux (Attacker Machine), Ubuntu 24.04 LTS (Wazuh Manager), and Ubuntu 22.04 LTS (Wazuh Agent).

Set up two network adapters for virtual machines: a host-only adapter for internal communication and a network address translation (NAT) for internet access. Ping was used to confirm connectivity between machines.

5.2 Installation of Wazuh Manager (Ubuntu 24.04)

Updated the system packages: `sudo apt update && sudo apt upgrade -y`.

Installed curl `sudo apt install curl apt-transport-https unzip wget -y` to fetch and download the official Wazuh installation script and required files directly from the Wazuh repository to the Ubuntu machine.

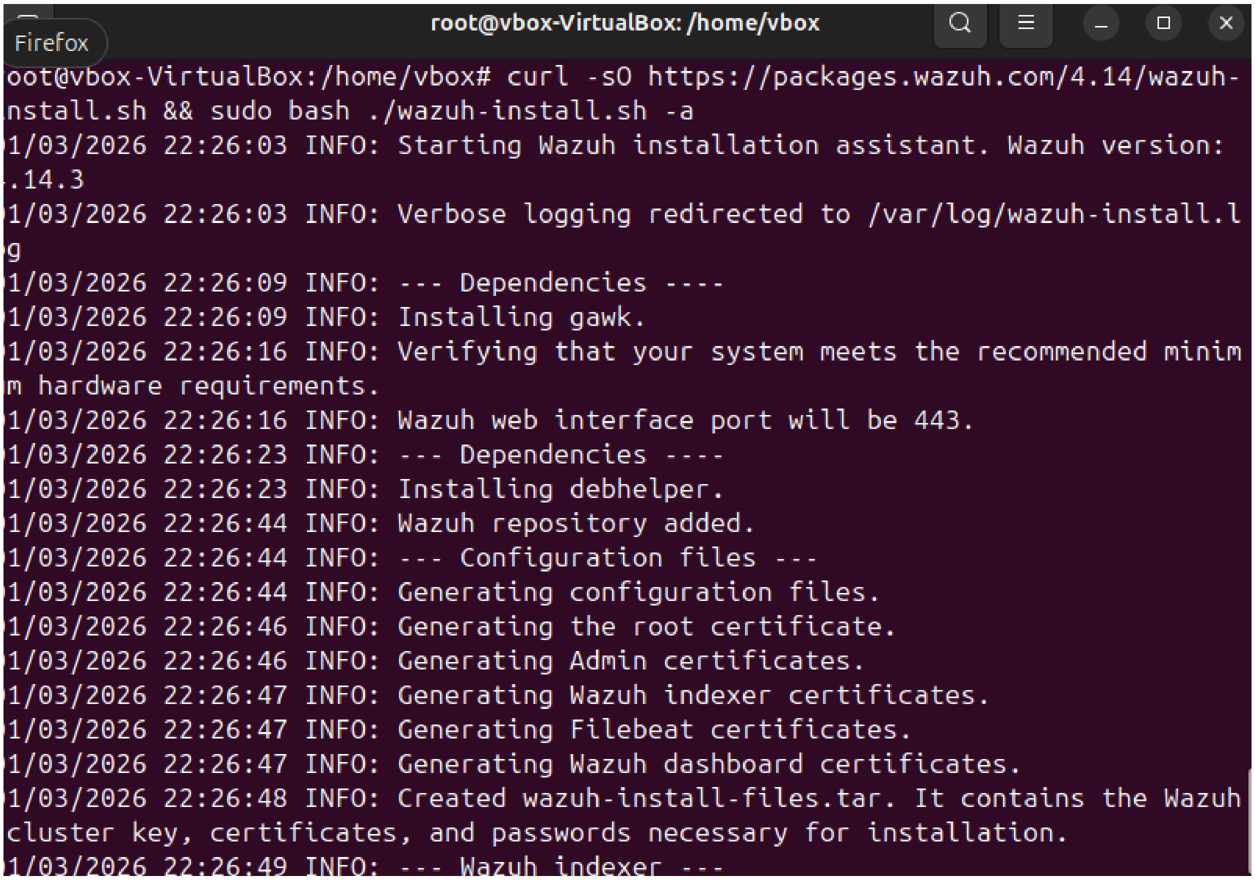


5.2.1. Update system package & install curl

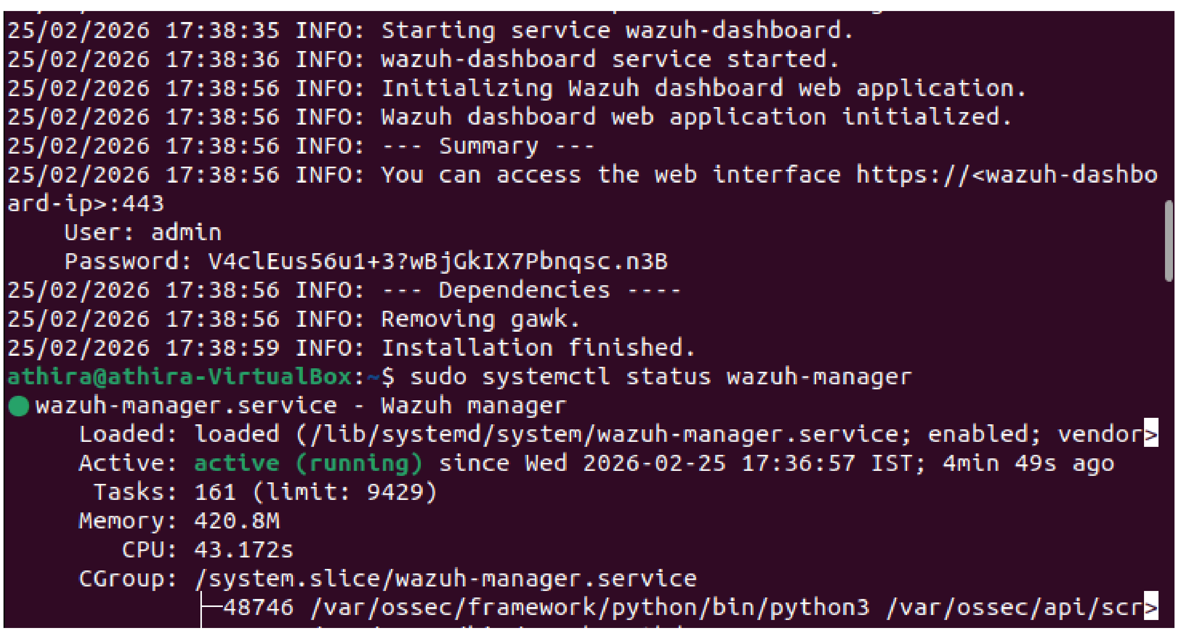
Installed Wazuh Manager (4.14), Wazuh Indexer, and Wazuh Dashboard on Ubuntu 24.04 LTS by downloading and running the official Wazuh installation script.

```
curl -sO https://packages.wazuh.com/4.14/wazuh-install.sh && sudo bash ./wazuh-install.sh -a
```

This output shows the access credentials and a message that confirms that the installation was successful.

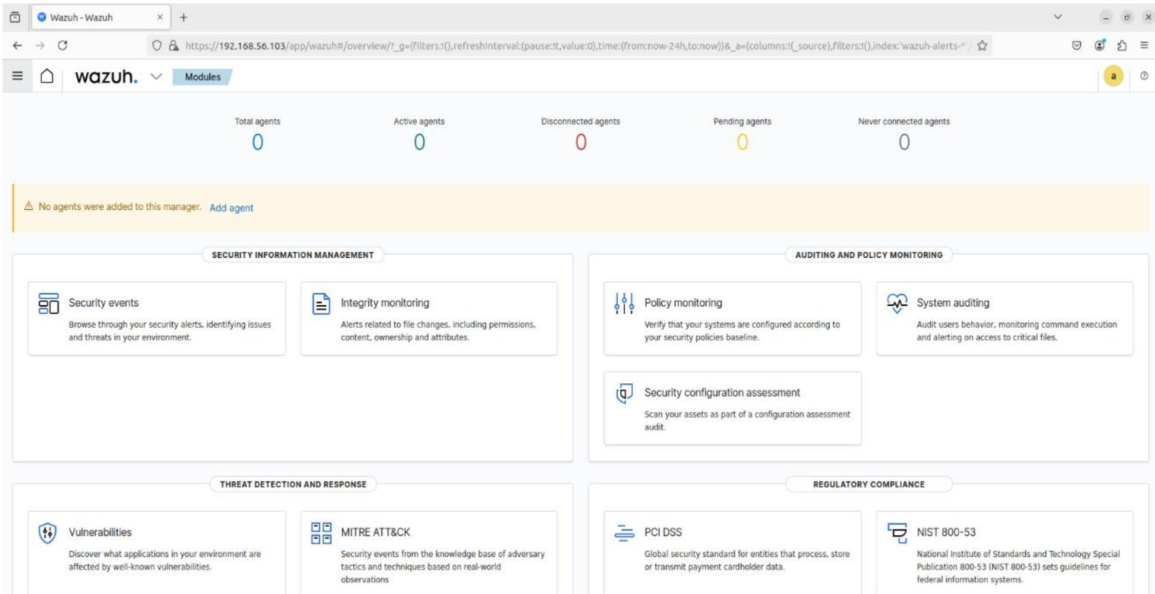


5.2.2. Downloading & Installing Wazuh Manager

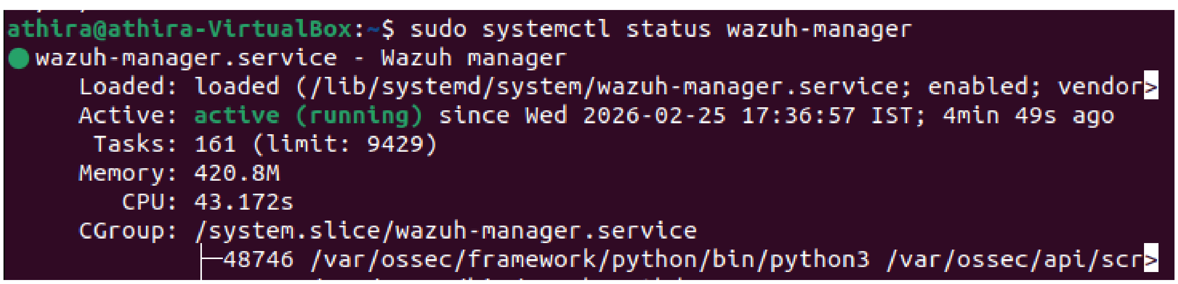


5.2.3. Credentials to access Wazuh Dashboard

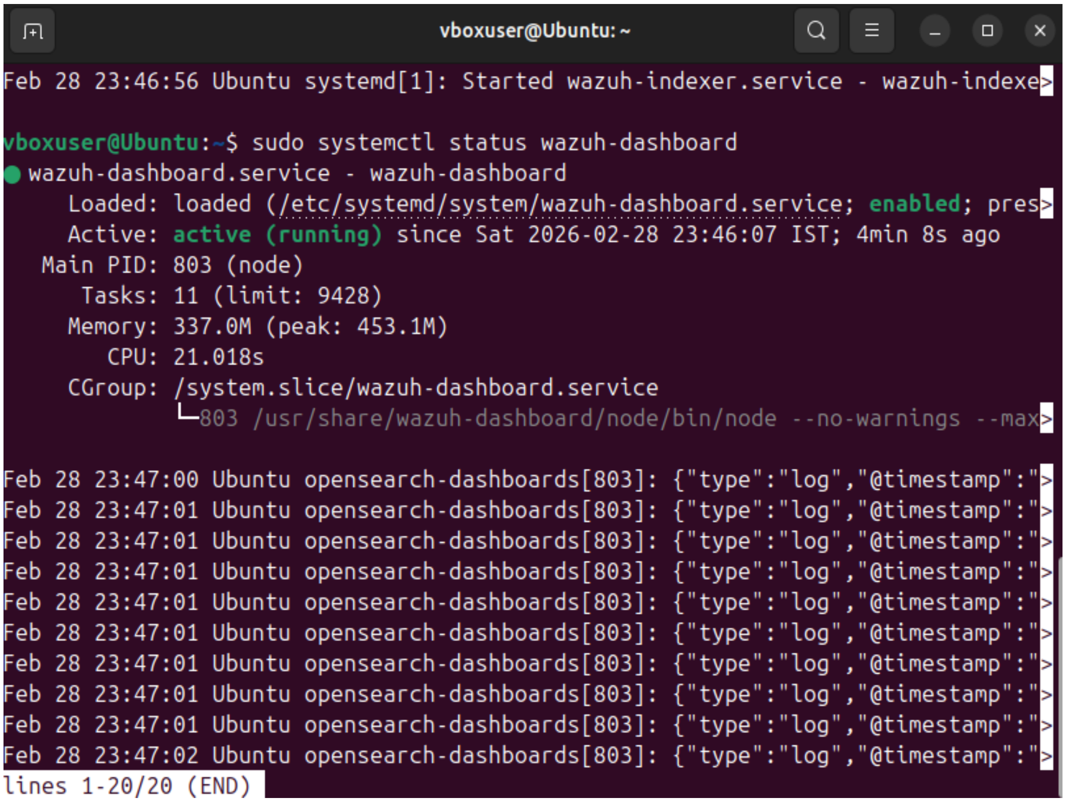
Verified that all services were running properly and accessed the Wazuh dashboard via browser using the Wazuh manager IP address.



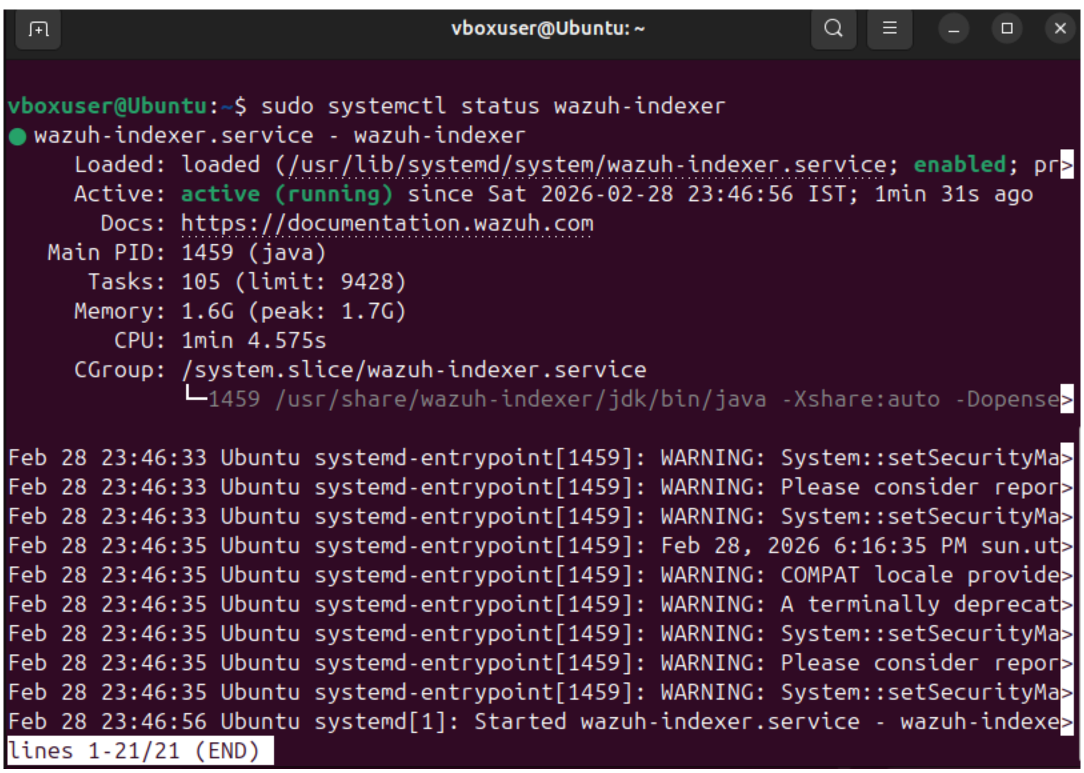
5.2.4. Wazuh Dashboard



5.2.5. Wazuh Manager status



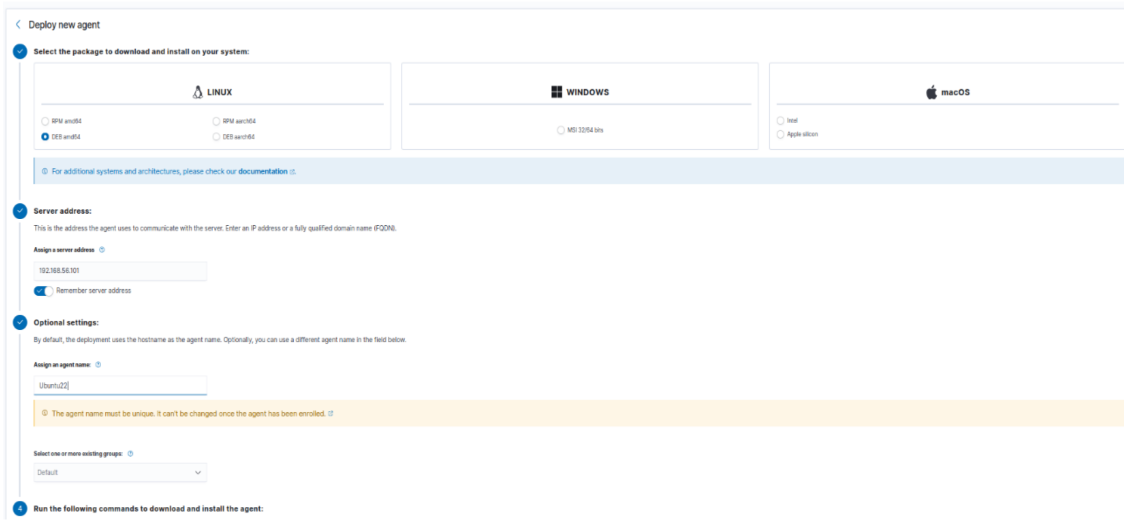
5.2.5. Wazuh Dashboard Status



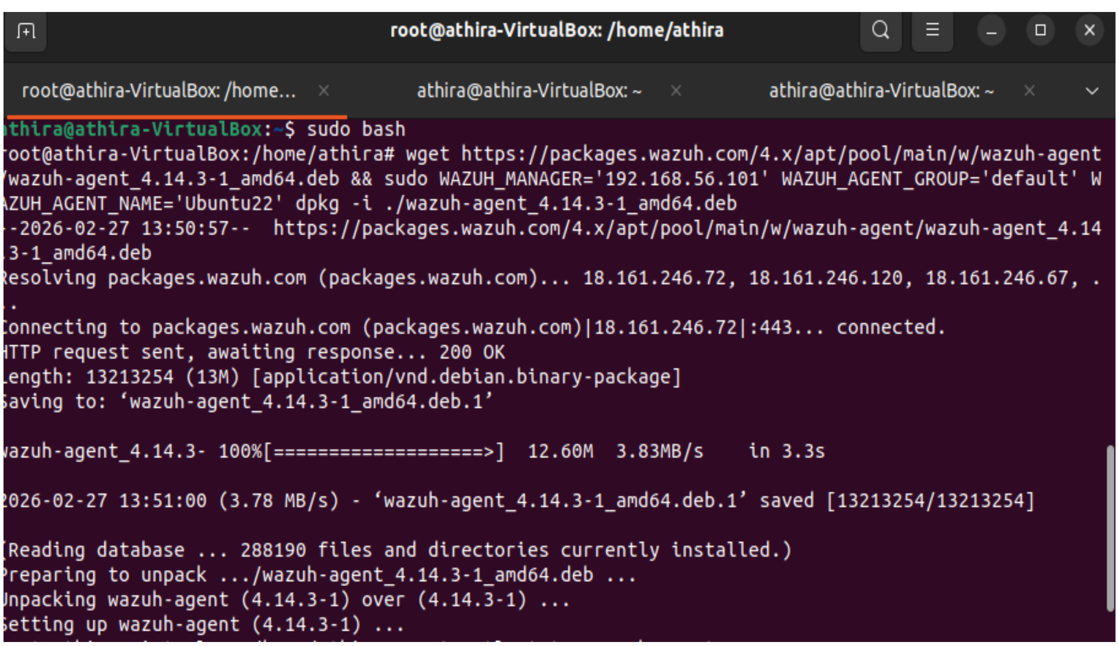
5.2.6. Wazuh-Indexer Status

5.3 Installation and Configuration of Wazuh Agent (Ubuntu 22.04)

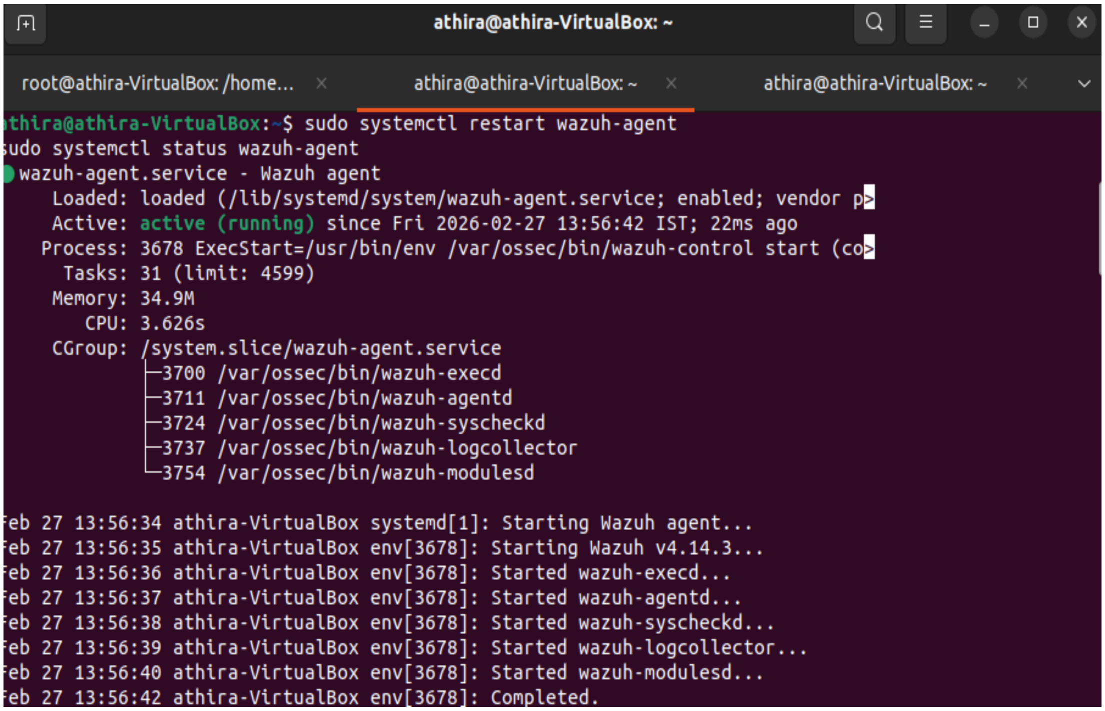
A new agent was created through the Wazuh interface. After that, the Wazuh agent was installed and configured on Ubuntu 22.04. Lately, the agent was registered and connected to the Wazuh Manager by configuring it with the Manager's IP address. Verified Wazuh-agent connection in Wazuh dashboard.



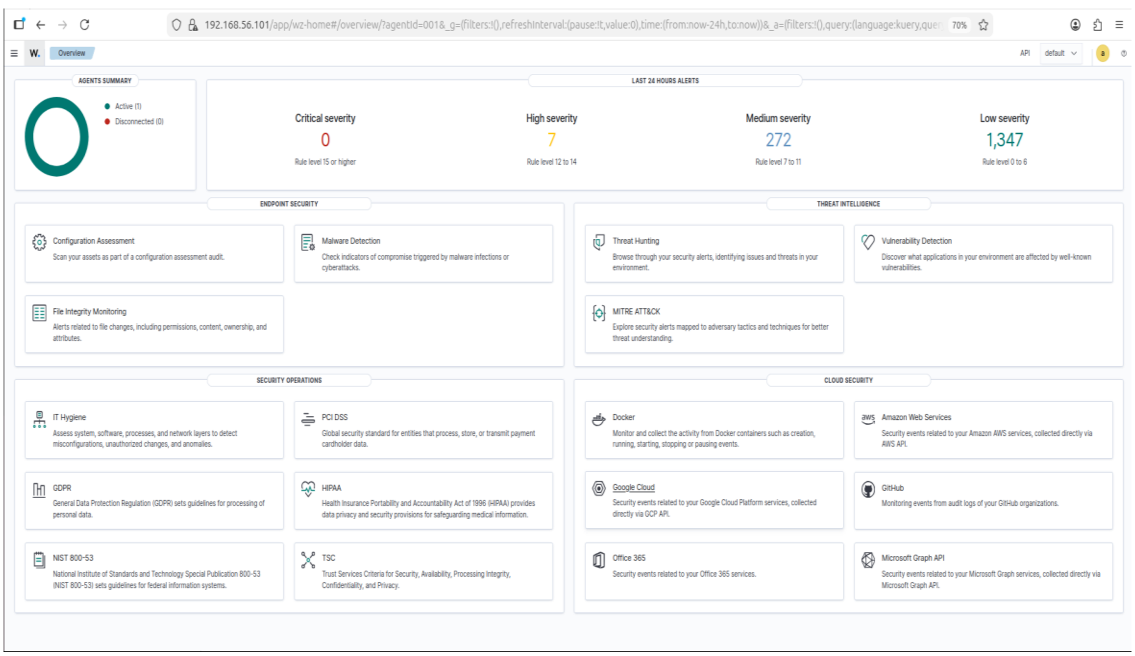
4.3.1 New Wazuh Agent Creation



4.3.2 Downloading & Installing Wazuh Agent



4.3.3. Wazuh-Agent Status



4.3.4. Wazuh Agent in the Wazuh Dashboard.

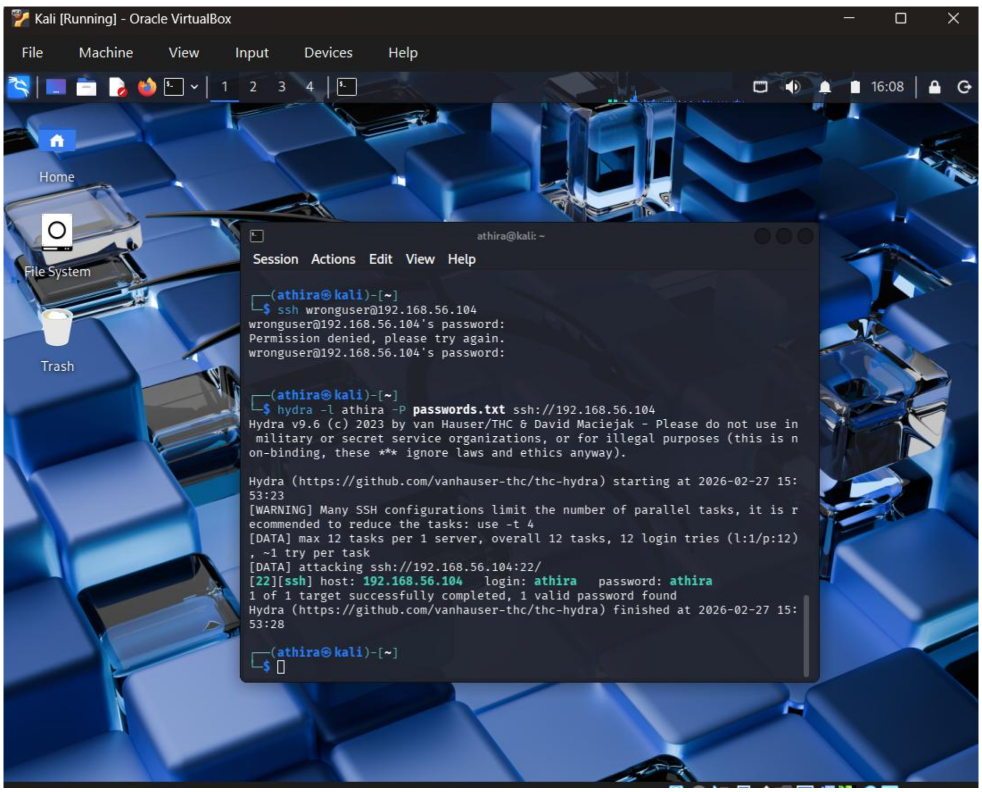
And also ensured effective security event collection and analysis. Authentication logs from /var/log/auth.log were verified, and system logs including SSH and sudo activities were confirmed to be properly recorded and forwarded to the manager. Real-time event visibility was validated through the Wazuh Dashboard.

5.4 Attack Simulation

Attack simulations were conducted from Kali Linux to generate security events and tried 4 different types of attacks:

SSH Brute Force Attack

- Created a password list file “password.txt” and used Hydra to attempt multiple failed SSH logins from Kali Linux to Ubuntu 22.04 Machine where agent was installed.



5.4.1. Brute Force Attack

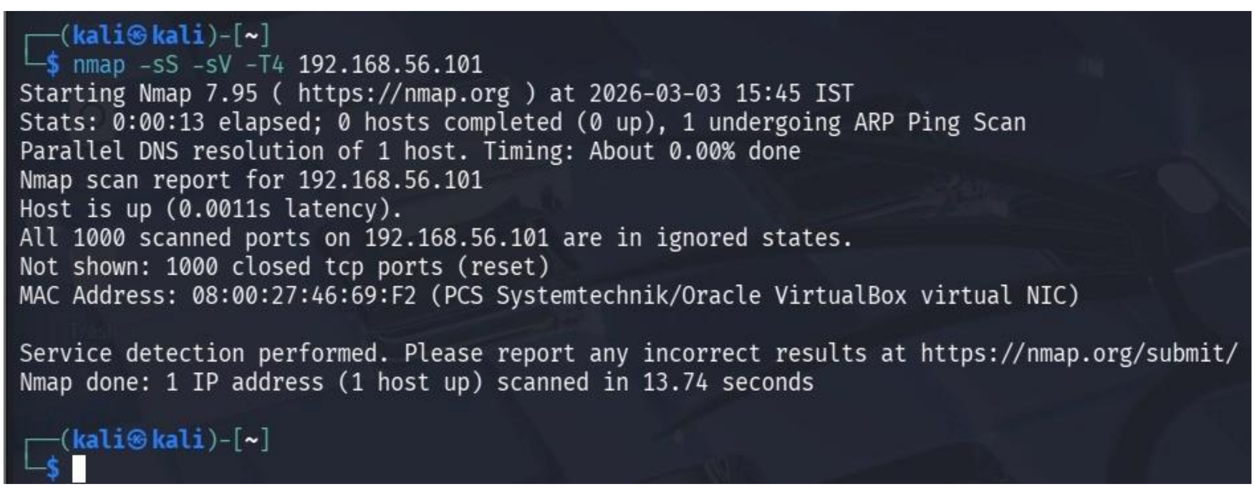
File Integrity / Modification Attack

- Modified /etc/passwd file of Ubuntu 22.04 Machine and generated the alert in Wazuh Dashboard.

5.4.2. File Modification

Port Scanning / Reconnaissance Attack

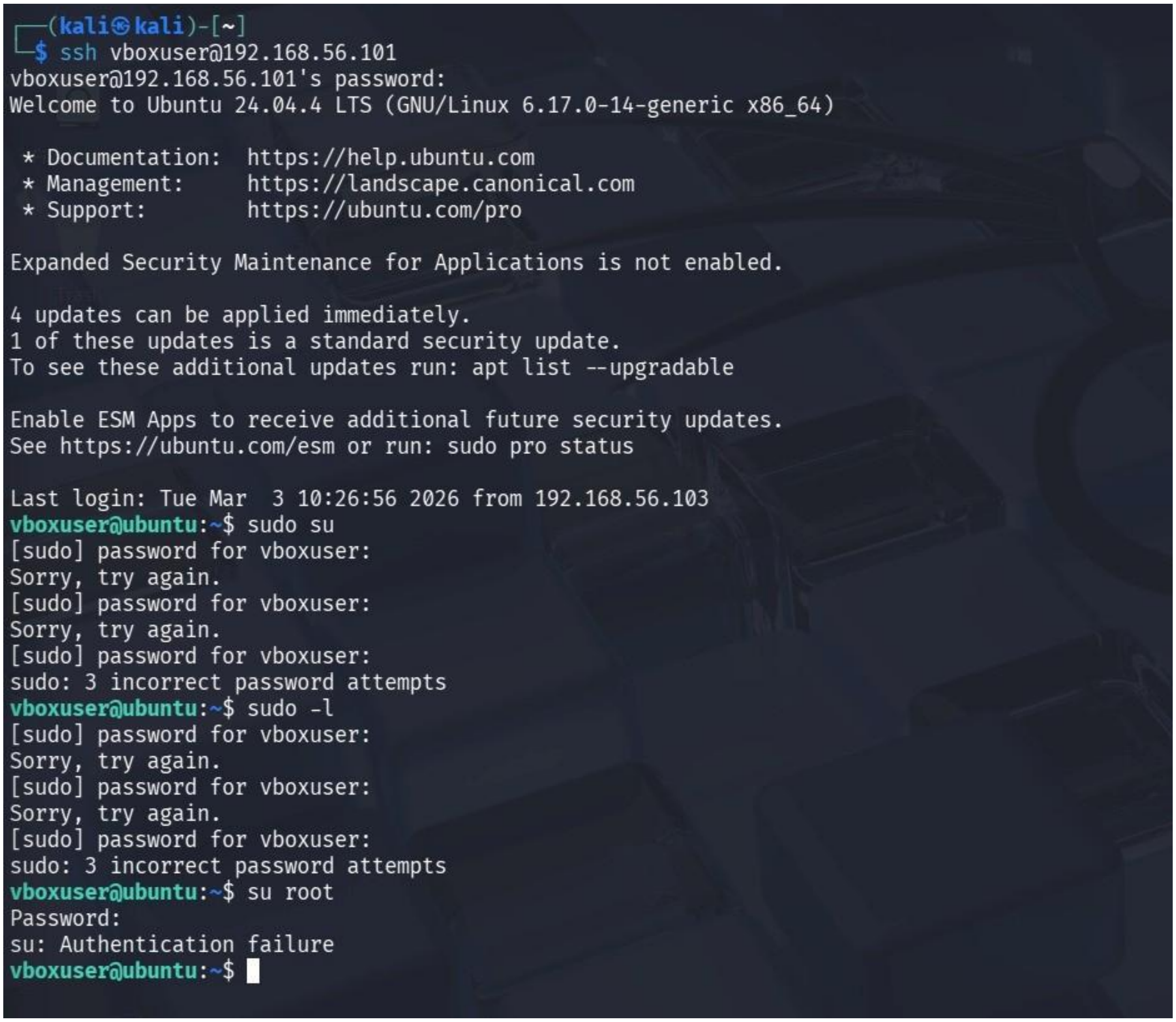
- Performed an Nmap port scan against the Ubuntu 22.04 machine and successfully generated reconnaissance alerts in the Wazuh Dashboard.



5.4.3. Port Scanning Attack

Privilege Escalation Attempt

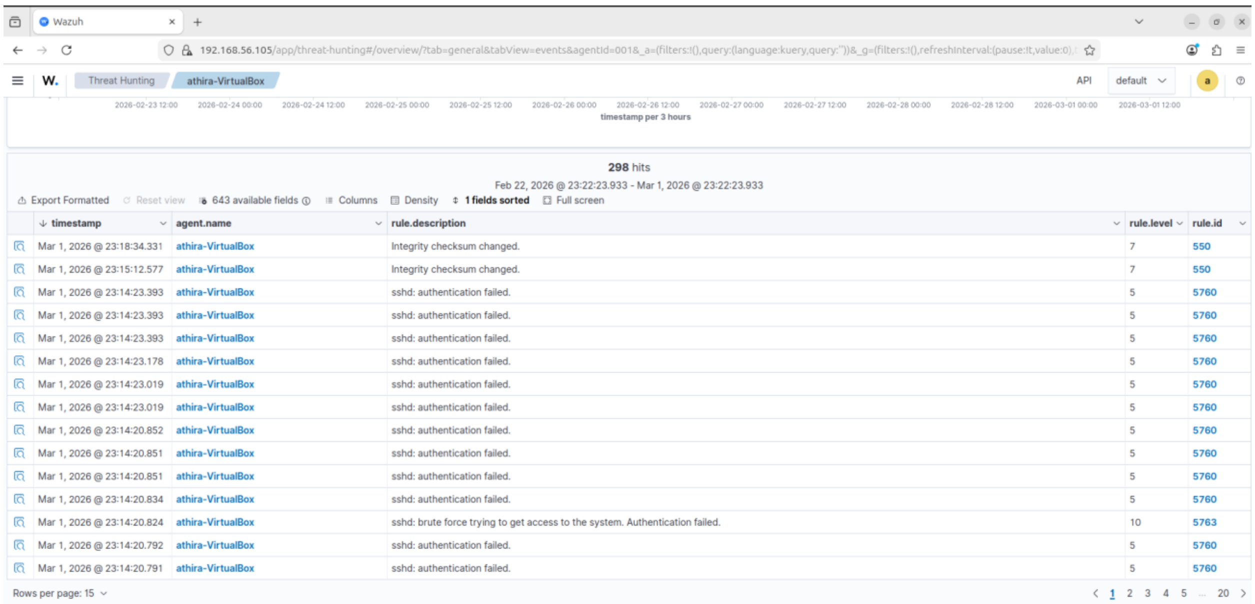
- Simulated a privilege escalation attempt on the Ubuntu 22.04 machine and successfully triggered authentication and sudo-related alerts in the Wazuh Dashboard.



5.4.4. Privilege Escalation Attempt

5.5 Log Collection and Alert Generation

In this phase Wazuh agent forwarded logs to the manager and manager analyzed logs using predefined rules. Therefore, alerts were triggered based on suspicious behavior.



timestamp	agent.name	rule.description	rule.level	rule.id
Mar 1, 2026 @ 23:18:34.331	athira-VirtualBox	Integrity checksum changed.	7	550
Mar 1, 2026 @ 23:15:12.577	athira-VirtualBox	Integrity checksum changed.	7	550
Mar 1, 2026 @ 23:14:23.393	athira-VirtualBox	sshd: authentication failed.	5	5760
Mar 1, 2026 @ 23:14:23.393	athira-VirtualBox	sshd: authentication failed.	5	5760
Mar 1, 2026 @ 23:14:23.393	athira-VirtualBox	sshd: authentication failed.	5	5760
Mar 1, 2026 @ 23:14:23.178	athira-VirtualBox	sshd: authentication failed.	5	5760
Mar 1, 2026 @ 23:14:23.019	athira-VirtualBox	sshd: authentication failed.	5	5760
Mar 1, 2026 @ 23:14:23.019	athira-VirtualBox	sshd: authentication failed.	5	5760
Mar 1, 2026 @ 23:14:20.852	athira-VirtualBox	sshd: authentication failed.	5	5760
Mar 1, 2026 @ 23:14:20.851	athira-VirtualBox	sshd: authentication failed.	5	5760
Mar 1, 2026 @ 23:14:20.851	athira-VirtualBox	sshd: authentication failed.	5	5760
Mar 1, 2026 @ 23:14:20.834	athira-VirtualBox	sshd: authentication failed.	5	5760
Mar 1, 2026 @ 23:14:20.824	athira-VirtualBox	sshd: brute force trying to get access to the system. Authentication failed.	10	5763
Mar 1, 2026 @ 23:14:20.792	athira-VirtualBox	sshd: authentication failed.	5	5760
Mar 1, 2026 @ 23:14:20.791	athira-VirtualBox	sshd: authentication failed.	5	5760

5.5.1. Brute Force Alerts

5.5.2 Raw Events of Brute Force Attempt

Document Details

View surrounding documents

View single document

t	location	journald
t	manager.name	vbox-VirtualBox
t	predecoder.hostname	athira-VirtualBox
t	predecoder.program_name	sshd
t	predecoder.timestamp	Mar 01 17:44:19
t	previous_output	Mar 1 23:14:19 athira-VirtualBox sshd[5844]: Failed password for athira from 192.168.56.102 port 43892 ssh2 Mar 01 17:44:19 athira-VirtualBox sshd[5845]: Failed password for athira from 192.168.56.102 port 43896 ssh2 Mar 1 23:14:19 athira-VirtualBox sshd[5843]: Failed password for athira from 192.168.56.102 port 43880 ssh2 Mar 1 23:14:19 athira-VirtualBox sshd[5841]: Failed password for athira from 192.168.56.
t	rule.description	sshd: brute force trying to get access to the system. Authentication failed.
#	rule.firedtimes	1
#	rule.frequency	8
t	rule.gdpr	IV_35.7.d, IV_32.2
t	rule.groups	syslog, sshd, authentication_failures
t	rule.hipaa	164.312.b
t	rule.id	5763
#	rule.level	10
	rule.mail	false
t	rule.mitre.id	T1110
t	rule.mitre.tactic	Credential Access
t	rule.mitre.technique	Brute Force
t	rule.nist_800_53	SI.4, AU.14, AC.7
t	rule.pci_dss	11.4, 10.2.4, 10.2.5

5.5.3 Details of Brute Force Alert

Document Details

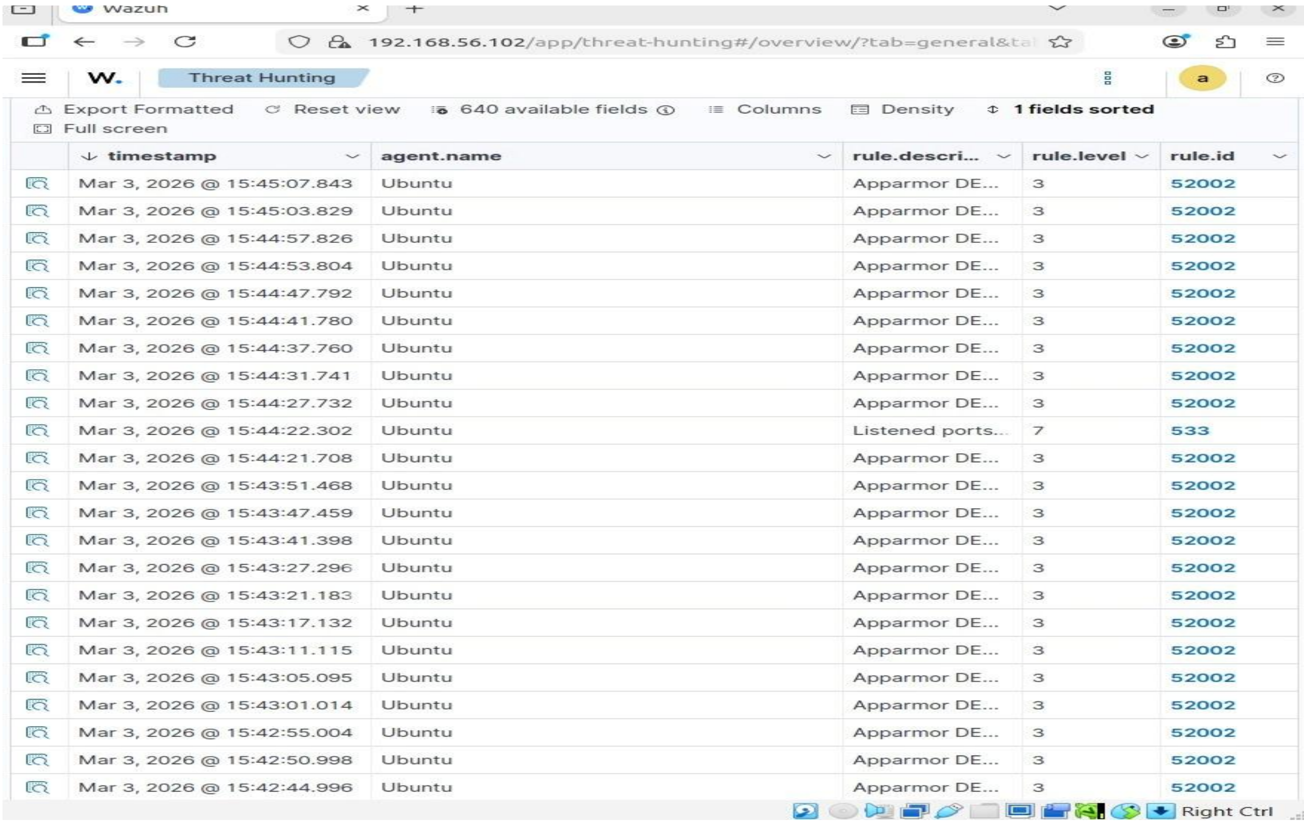
View surrounding documents

View single document

Table JSON

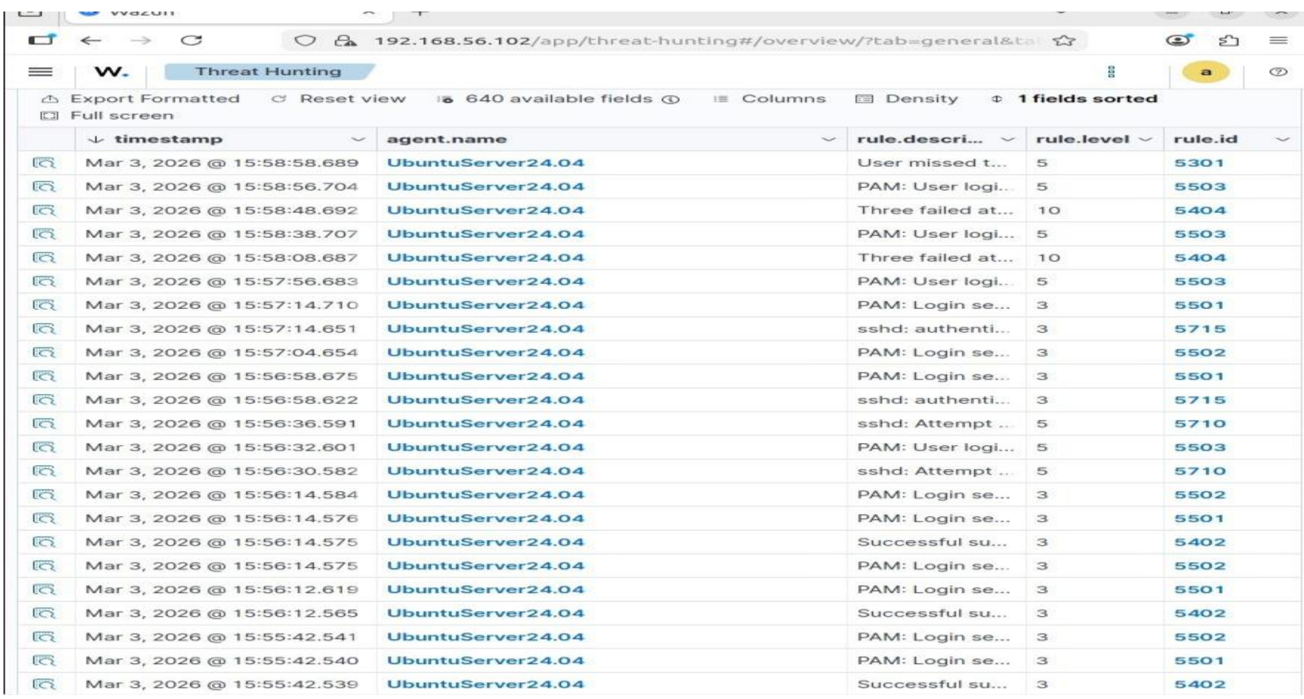
t	_index	wazuh-alerts-4.x-2026.02.27
t	agent.id	001
t	agent.ip	192.168.56.104
t	agent.name	athira-VirtualBox
t	decoder.name	syscheck_integrity_changed
t	full_log	File '/etc/passwd' modified Mode: scheduled Changed attributes: size,mtime,md5,sha1,sha256 Size changed from '3130' to '3124' Old modification time was: '1772192352', now it is '1772192767' Old md5sum was: '23ead6f3baeec3393a099e4e5721fb26' New md5sum is : '0d54a411cb7294c1de4adc1e5e289a3c'
t	id	1772192802.2703865
t	input.type	log
t	location	syscheck
t	manager.name	Ubuntu
t	rule.description	Integrity checksum changed.
#	rule.firedtimes	1
t	rule.gdpr	II_5.1.f
t	rule.gpg13	4.11
t	rule.groups	ossec, syscheck, syscheck_entry_modified, syscheck_file
t	rule.hipaa	164.312.c.1, 164.312.c.2
t	rule.id	550
#	rule.level	7
	rule.mail	false
t	rule.mitre.id	T1565.001
t	rule.mitre.tactic	Impact
t	rule.mitre.technique	Stored Data Manipulation
t	rule.nist_800_53	SI.7

5.5.4. Integrity Checksum changed Alert



timestamp	agent.name	rule.descri...	rule.level	rule.id
Mar 3, 2026 @ 15:45:07.843	Ubuntu	Apparmor DE...	3	52002
Mar 3, 2026 @ 15:45:03.829	Ubuntu	Apparmor DE...	3	52002
Mar 3, 2026 @ 15:44:57.826	Ubuntu	Apparmor DE...	3	52002
Mar 3, 2026 @ 15:44:53.804	Ubuntu	Apparmor DE...	3	52002
Mar 3, 2026 @ 15:44:47.792	Ubuntu	Apparmor DE...	3	52002
Mar 3, 2026 @ 15:44:41.780	Ubuntu	Apparmor DE...	3	52002
Mar 3, 2026 @ 15:44:37.760	Ubuntu	Apparmor DE...	3	52002
Mar 3, 2026 @ 15:44:31.741	Ubuntu	Apparmor DE...	3	52002
Mar 3, 2026 @ 15:44:27.732	Ubuntu	Apparmor DE...	3	52002
Mar 3, 2026 @ 15:44:22.302	Ubuntu	Listened ports...	7	533
Mar 3, 2026 @ 15:44:21.708	Ubuntu	Apparmor DE...	3	52002
Mar 3, 2026 @ 15:43:51.468	Ubuntu	Apparmor DE...	3	52002
Mar 3, 2026 @ 15:43:47.459	Ubuntu	Apparmor DE...	3	52002
Mar 3, 2026 @ 15:43:41.398	Ubuntu	Apparmor DE...	3	52002
Mar 3, 2026 @ 15:43:27.296	Ubuntu	Apparmor DE...	3	52002
Mar 3, 2026 @ 15:43:21.183	Ubuntu	Apparmor DE...	3	52002
Mar 3, 2026 @ 15:43:17.132	Ubuntu	Apparmor DE...	3	52002
Mar 3, 2026 @ 15:43:11.115	Ubuntu	Apparmor DE...	3	52002
Mar 3, 2026 @ 15:43:05.095	Ubuntu	Apparmor DE...	3	52002
Mar 3, 2026 @ 15:43:01.014	Ubuntu	Apparmor DE...	3	52002
Mar 3, 2026 @ 15:42:55.004	Ubuntu	Apparmor DE...	3	52002
Mar 3, 2026 @ 15:42:50.998	Ubuntu	Apparmor DE...	3	52002
Mar 3, 2026 @ 15:42:44.996	Ubuntu	Apparmor DE...	3	52002

5.5.5. Reconnaissance Alerts

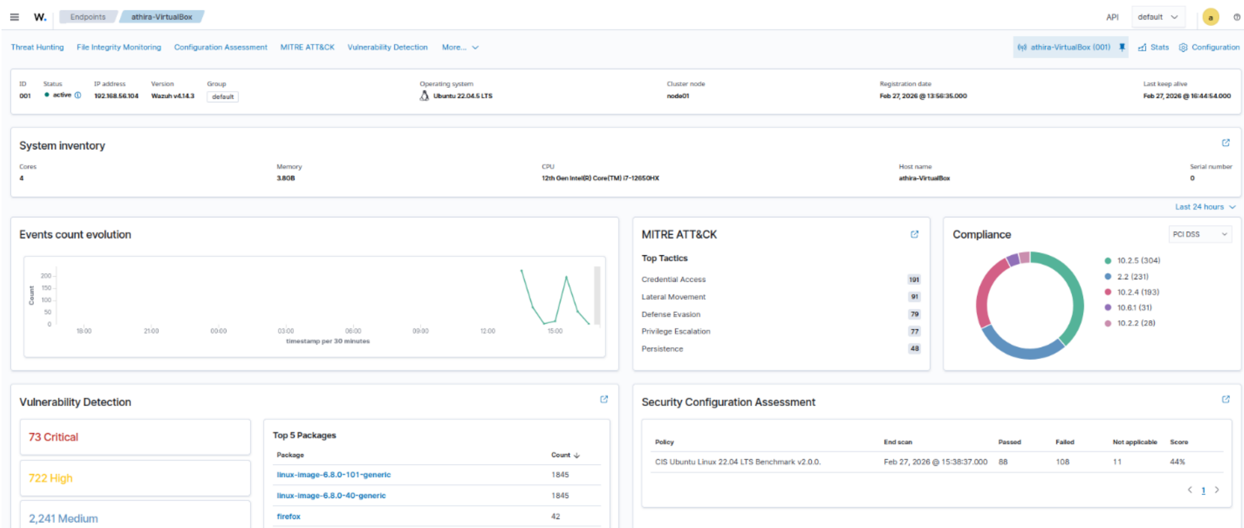


timestamp	agent.name	rule.descri...	rule.level	rule.id
Mar 3, 2026 @ 15:58:58.689	UbuntuServer24.04	User missed t...	5	5301
Mar 3, 2026 @ 15:58:56.704	UbuntuServer24.04	PAM: User logi...	5	5503
Mar 3, 2026 @ 15:58:48.692	UbuntuServer24.04	Three failed at...	10	5404
Mar 3, 2026 @ 15:58:38.707	UbuntuServer24.04	PAM: User logi...	5	5503
Mar 3, 2026 @ 15:58:08.687	UbuntuServer24.04	Three failed at...	10	5404
Mar 3, 2026 @ 15:57:56.683	UbuntuServer24.04	PAM: User logi...	5	5503
Mar 3, 2026 @ 15:57:14.710	UbuntuServer24.04	PAM: Login se...	3	5501
Mar 3, 2026 @ 15:57:14.651	UbuntuServer24.04	sshd: authenti...	3	5715
Mar 3, 2026 @ 15:57:04.654	UbuntuServer24.04	PAM: Login se...	3	5502
Mar 3, 2026 @ 15:56:58.675	UbuntuServer24.04	PAM: Login se...	3	5501
Mar 3, 2026 @ 15:56:58.622	UbuntuServer24.04	sshd: authenti...	3	5715
Mar 3, 2026 @ 15:56:36.591	UbuntuServer24.04	sshd: Attempt ...	5	5710
Mar 3, 2026 @ 15:56:32.601	UbuntuServer24.04	PAM: User logi...	5	5503
Mar 3, 2026 @ 15:56:30.582	UbuntuServer24.04	sshd: Attempt ...	5	5710
Mar 3, 2026 @ 15:56:14.584	UbuntuServer24.04	PAM: Login se...	3	5502
Mar 3, 2026 @ 15:56:14.576	UbuntuServer24.04	PAM: Login se...	3	5501
Mar 3, 2026 @ 15:56:14.575	UbuntuServer24.04	Successful su...	3	5402
Mar 3, 2026 @ 15:56:14.575	UbuntuServer24.04	PAM: Login se...	3	5502
Mar 3, 2026 @ 15:56:12.619	UbuntuServer24.04	PAM: Login se...	3	5501
Mar 3, 2026 @ 15:56:12.565	UbuntuServer24.04	Successful su...	3	5402
Mar 3, 2026 @ 15:55:42.541	UbuntuServer24.04	PAM: Login se...	3	5502
Mar 3, 2026 @ 15:55:42.540	UbuntuServer24.04	PAM: Login se...	3	5501
Mar 3, 2026 @ 15:55:42.539	UbuntuServer24.04	Successful su...	3	5402

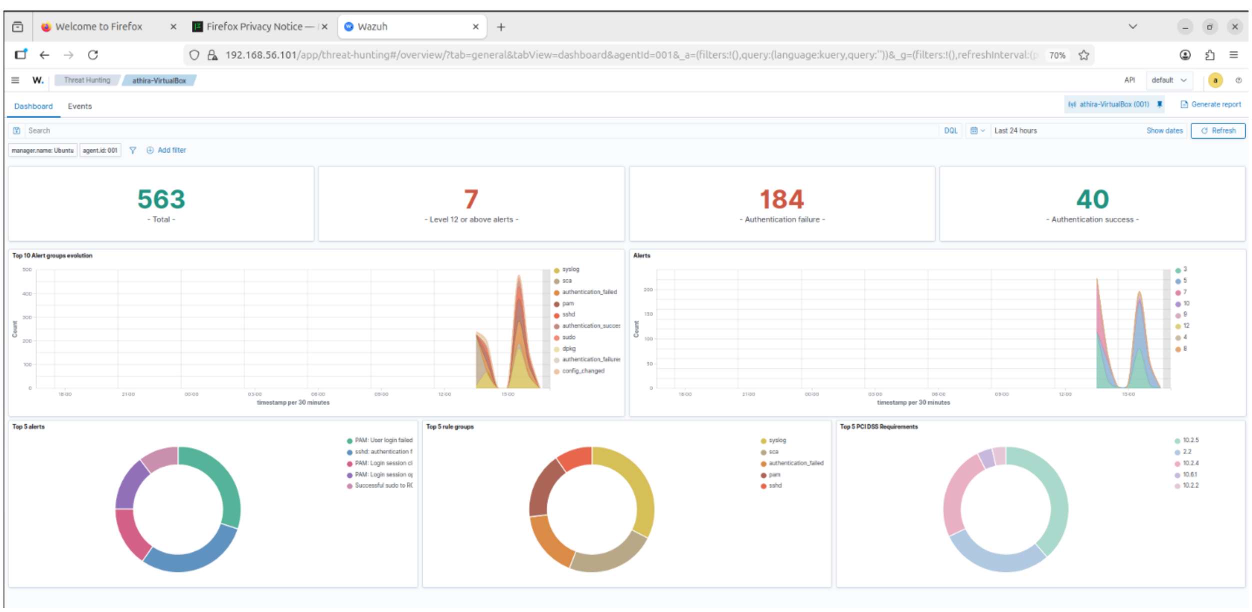
5.5.6. Privilege Escalation Alerts

5.6 Dashboard Analysis and Visualization

The status of the connected machine and the security alerts it has generated are displayed on the Wazuh Agent default dashboard. It shows basic information such as the operating system, IP address, and system name. It helps monitor activities and detect suspicious actions on the endpoint machine.

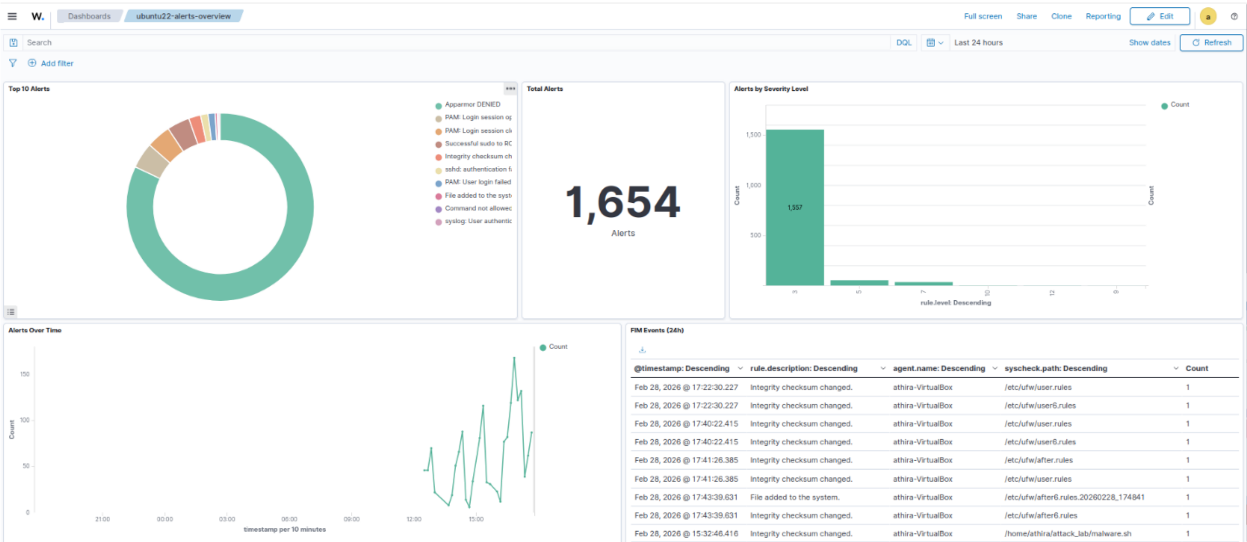


5.6.1. Agent-Dashboard



5.6.2. Agent-Thread Hunting Dashboard

A customized dashboard was created based on the last 24 hours of data, including the Top 10 alerts, total alert count, alerts by severity level, file integrity monitoring (FIM) events, and alerts over a time period. The dashboard was also saved under the name ‘ubuntu24_alerts_overview’ for future reference.



5.6.3. Customized Dashboard for Ubuntu 22 Machine

6. Challenges

- One of the challenges in the lab setup was ensuring compatibility between Ubuntu OS version and Wazuh server version.
- Wazuh components require high memory and processing power, especially in a virtual machine environment, which leads to slow performance and occasional service issues.

7. Conclusion

This project demonstrated the setup and implementation of a basic security monitoring lab environment. In a controlled virtual environment, a centralized monitoring system was put up to gather logs, identify suspicious activity, and produce security alerts.

The project provided hands-on experience in log analysis, alert generation, and security event monitoring. During the setup procedure, a number of technical difficulties were encountered, such as system compatibility issues and configuration concerns. Addressing these issues helped improve troubleshooting abilities and understanding of how monitoring systems operate in real-world environments.

The project highlighted the importance of continuous monitoring, proper configuration, and timely detection of potential security incidents. All things considered, this lab experience enhanced fundamental understanding of security monitoring and incident detection while developing useful abilities relevant to entry-level cybersecurity and SOC positions.

8. References

1. https://documentation.wazuh.com/current/getting-started/index.html
2. https://youtu.be/cH4psUe-7P4?si=ZIPZUCahYYPYdWLv
3. https://www.youtube.com/watch?v=awOZ89VF56Y&t=3s
4. https://www.youtube.com/watch?v=QrcAhd5P7xw
5. https://www.geeksforgeeks.org/ethical-hacking/introduction-to-wazuh